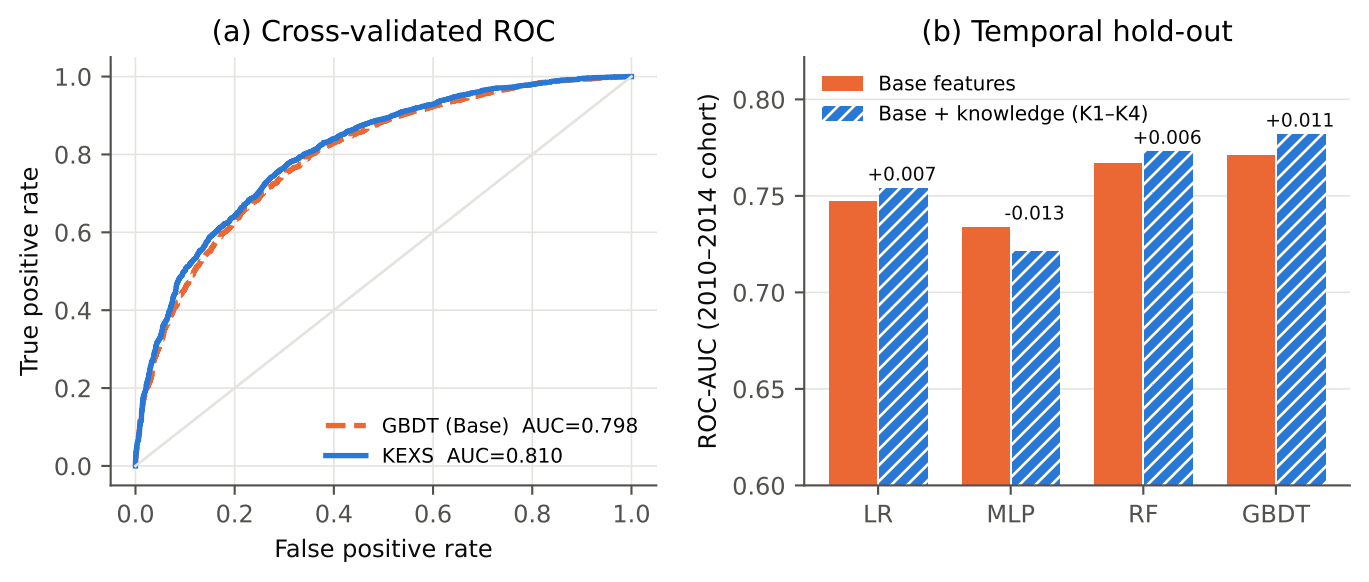
(a) Cross-validated ROC
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
GBDT (Base) AUC=0.798
KEXS AUC=0.810
(b) Temporal hold-out
ROC-AUC (2010–2014 cohort)
0.60
0.65
0.70
0.75
0.80
Base features
Base + knowledge (K1–K4)
+0.007
-0.013
+0.006
+0.011
LR
MLP
RF
GBDT